

Statistical and Computational Developments in Osteoarthritis Research

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Basic Science Lecture Series

London, ON

July 22, 2026



Osteoarthritis (OA)

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HEALTH

Osteoarthritis cases could reach 1 billion worldwide by 2050, study says



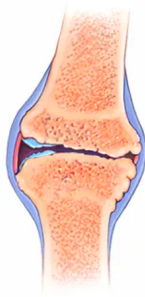
By Eric Stober • Global News

Posted August 21, 2023 5:30 pm • Updated August 22, 2023 7:58 am • 2 min read

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Healthy joint



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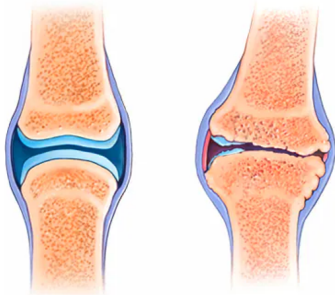
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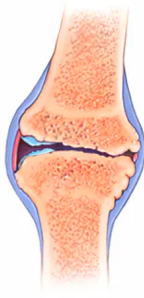


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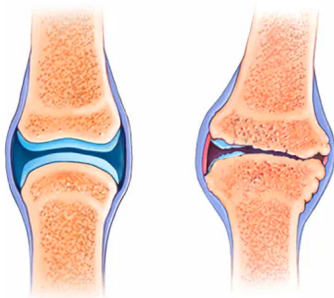
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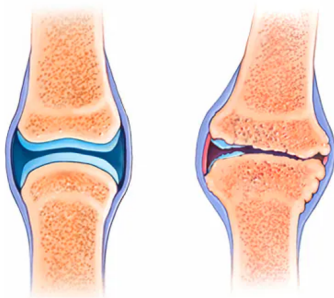
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 $\Rightarrow$ Prevention and Treatment.
- Demographic, imaging, biospecimen information available.
- Three subcohorts: Progression ($\approx 29\%$), Incidence ($\approx 68\%$), and Control ($\approx 3\%$).

A latent class linear mixed model for monotonic continuous processes measured with error

Oswaldo Espin-García^{1,2,3,4,*} , Lizbeth Naranjo^{5,*} 
and Ruth Fuentes-García⁵

Statistical Methods in Medical Research
1–16

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DOI: 10.1177/09622802231225963

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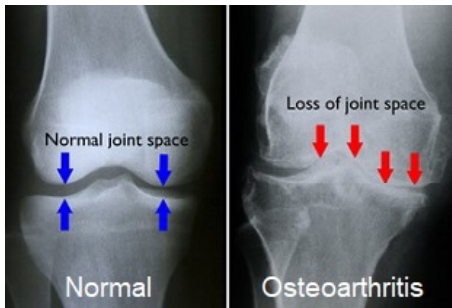
Abstract

Motivated by measurement errors in radiographic diagnosis of osteoarthritis, we propose a Bayesian approach to identify latent classes in a model with continuous response subject to a monotonic, that is, non-decreasing or non-increasing, process with measurement error. A latent class linear mixed model has been introduced to consider measurement error while the monotonic process is accounted for via truncated normal distributions. The main purpose is to classify the response trajectories through the latent classes to better describe the disease progression within homogeneous subpopulations.

Keywords

Bayesian analysis, disease trajectories, latent class linear mixed models, measurement error, monotonic continuous process

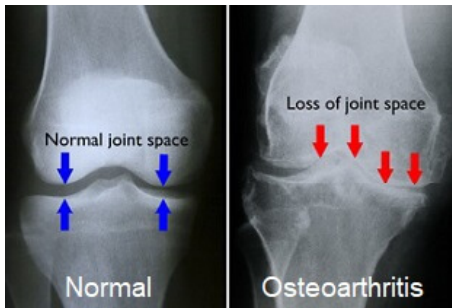
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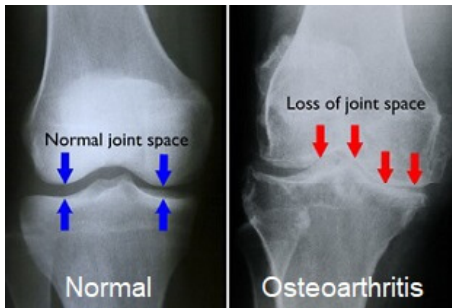
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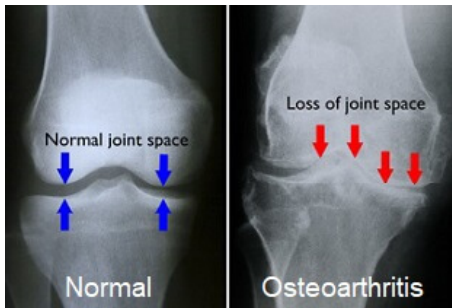
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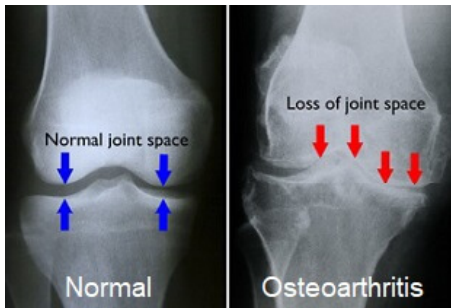
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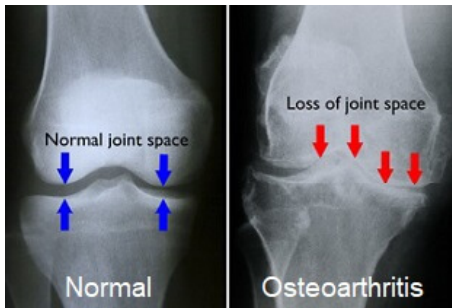
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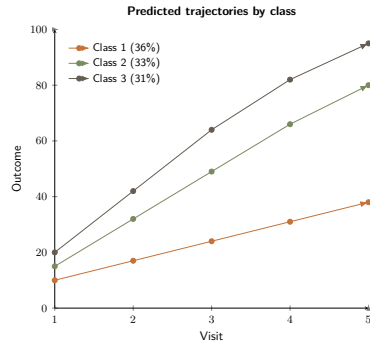
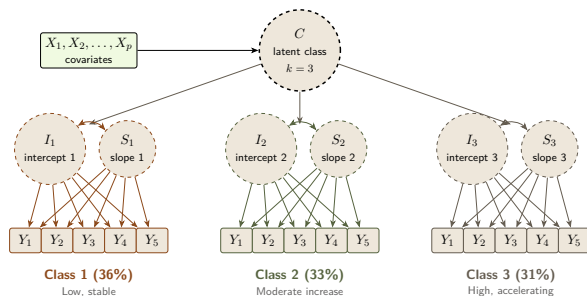


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Objective: To identify trajectory groups based on the reported medial minimum joint space width (MCMJSW) and available risk factors.

Latent Trajectory Models



Inference - Bayesian Analysis

- Posteriors obtained via Markov Chain Monte Carlo using JAGS.

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- Compared the proposed model against a “standard” LCLMM, i.e., without addressing measurement error nor monotonicity.

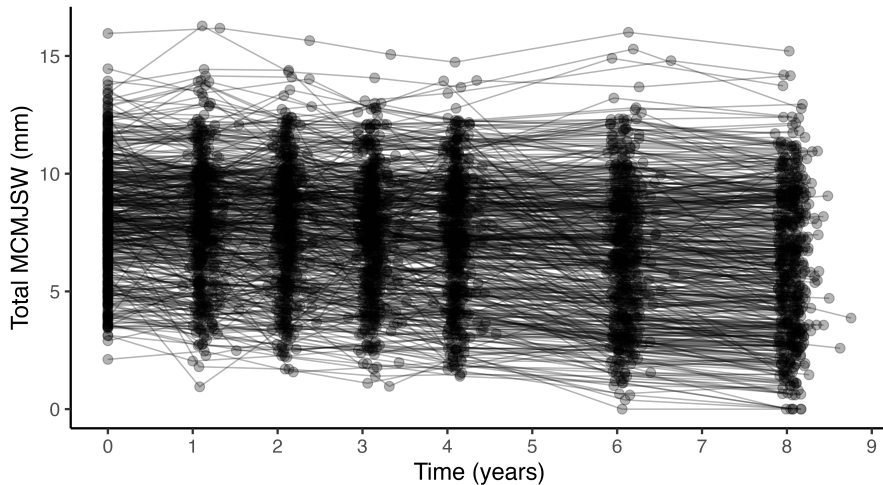
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- Compared the proposed model against a “standard” LCLMM, i.e., without addressing measurement error nor monotonicity.
- Simulations showed that the proposed model has improvements in model selection with increasing sample size and number of timepoints.

OAI - Baseline characteristics

Covariate	Progression (n=505)	Incidence (n=1084)	Control (n=75)
MCMJSWsum			
Mean (sd)	8.2 (2.3)	7.2 (2.6)	9.5 (1.6)
Median (Min,Max)	8.2 (2.1,16.0)	7.4 (0.0,15.7)	9.0 (6.7,13.7)
Sex			
F	270 (53)	633 (58)	46 (61)
M	235 (47)	451 (42)	29 (39)
Age			
Mean (sd)	59.7 (8.8)	61.3 (8.7)	54.3 (7.3)
Median (Min,Max)	59 (45,79)	61 (45,79)	53 (45,76)
BMI			
Mean (sd)	29.7 (4.7)	28.4 (4.6)	24.0 (2.9)
Median (Min,Max)	29.4 (18.2,44.2)	28.0 (17.6,44.3)	23.8 (17.7,31.4)
WOMAC TS			
Mean (sd)	23.5 (16.9)	10.2 (12.3)	1.0 (3.3)
Median (Min,Max)	20.9 (0.0,85.0)	6.0 (0.0,81.4)	0 (0,21)

Total JSW Trajectories (Progression subcohort)



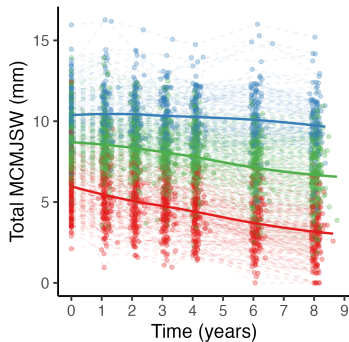
Results - Model Selection

- Chose the number of classes using Watanabe-Akaike Information Criterion (WAIC) and Leave-One-Out cross-validation (LOO)

K	WAIC		LOO	
	LCLMM	Proposal	LCLMM	Proposal
2	9920.87	8199.82	10411.07	8614.59
3	9735.32	6801.50	10222.50	7328.03
4	9770.05	6885.16	10219.89	7345.95
5	9724.57	6672.77	10192.98	7117.77

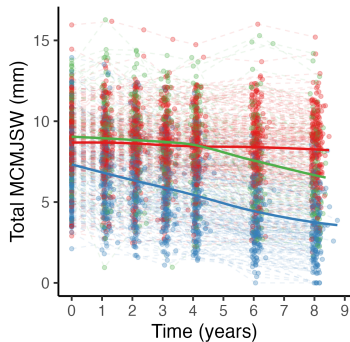
Results - Classes

ECR-1, K=3, and 30K MCMC iterations



Class

- 1 (32.1)
- 2 (23.8)
- 3 (44.2)

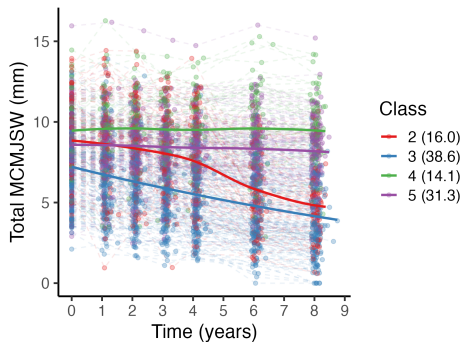
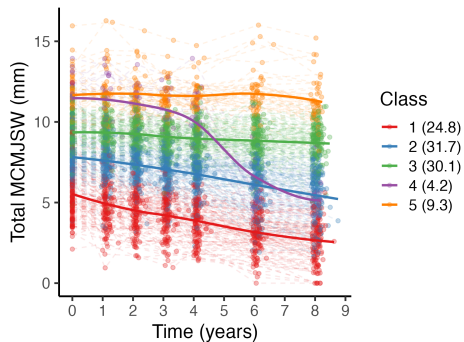


Class

- 1 (45.3)
- 2 (38.2)
- 3 (16.4)

Results - Classes

ECR-1, K=5, and 30K MCMC iterations



Results - Estimation

ECR-1, K=3, and 30K MCMC iterations

Param.	Variable	LCLMM			Proposal		
		Mean	SD	95% CI	Mean	SD	95% CI
β_0	Intercept	13.205	0.452	(12.3,14.1)	13.811	0.559	(12.6,14.8)
β	Age	-0.067	0.006	(-0.0781,-0.0559)	-0.073	0.007	(-0.0855,-0.0595)
	Sex (ref. F)	0.705	0.101	(0.502,0.903)	0.483	0.123	(0.244,0.725)
	BMI	-0.027	0.009	(-0.0435,-0.00931)	-0.016	0.012	(-0.0402,0.00878)
	WOMAC	-0.004	0.002	(-0.00653,-0.000481)	-0.021	0.003	(-0.028,-0.015)
Γ	Γ_{11}	1.0 constraint			1.0 constraint		
	$\Gamma_{21} = \Gamma_{12}$	0.000	0.005	(-0.0107,0.0107)	-0.002	0.020	(-0.0404,0.0377)
	Γ_{22}	0.027	0.003	(0.0224,0.0327)	0.064	0.014	(0.0399,0.0952)
λ_1	Intercept	0 constraint			0 constraint		
	Slope	-0.056	0.031	(-0.115,0.00349)	9.305	1.033	(7.56,11.6)
	Quadratic	-0.000	0.003	(-0.0065,0.00664)	-1.066	0.131	(-1.35,-0.841)
λ_2	Intercept	-1.272	0.124	(-1.52,-1.03)	-1.015	0.142	(-1.29,-0.734)
	Slope	-0.523	0.033	(-0.588,-0.459)	0.333	0.068	(0.196,0.468)
	Quadratic	0.008	0.004	(0.00149,0.0154)	-0.084	0.009	(-0.101,-0.067)
λ_3	Intercept	0.380	0.159	(0.0733,0.69)	0.852	0.213	(0.439,1.27)
	Slope	-0.011	0.051	(-0.11,0.0893)	1.724	0.316	(1.16,2.44)
	Quadratic	-0.036	0.006	(-0.0474,-0.0254)	-0.244	0.039	(-0.33,-0.175)
σ^2	Class ₁	No error			0.207	0.009	(0.191,0.224)
	Class ₂				0.190	0.014	(0.165,0.219)
	Class ₃				2.197	0.179	(1.88,2.56)
τ^2		0.715	0.022	(0.674,0.759)	1.0 constraint		

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- Bayesian methods allow for data complexities such as measurement error and monotonicity in a continuous process.
- Nonidentifiability is common in LCLMM, label switching is addressed via a post-processing approach that can aid to circumvent this issue but requires careful analysis of the problem.
- Intrinsic disease characteristics, e.g., chronic deterioration, can inform latent variable specifications.

P2 - Data-driven methods to quantify physical activity (PA) levels

In collaboration with S. Amanda Ali and Jackie L. Whittaker

SEDENTARY



At rest

LIGHT



No increase in
breathing/sweating

MODERATE



Slight increase in
breathing/sweating

VIGOROUS



Large increase in
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- **What is PA?**

“Any bodily movement caused by muscle contractions and energy expenditure”
Caspersen et al, 1985.

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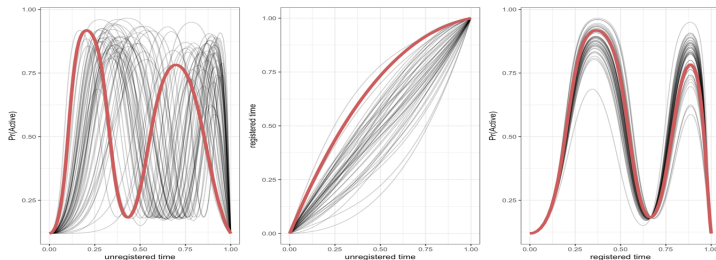
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- Wearable devices, e.g., accelerometers, offer a way to objectively measure PA frequency, duration and intensity.
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Objective: To utilize curve registration methods to improve the comparability of accelerometer-measured PA data.

Methods

- Curve registration methods (Wrobel et al, 2019) aim to facilitate comparison of PA by arranging data to a common time scale.
- Operates through a two-step process alternating between
 - ① an inverse warping function estimation to capture horizontal variability, and
 - ② functional principal components analysis to grasp vertical variability.



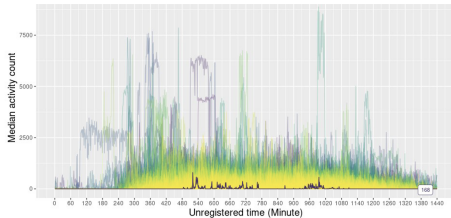
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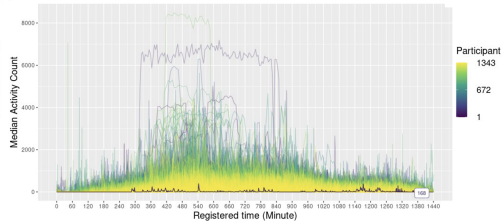
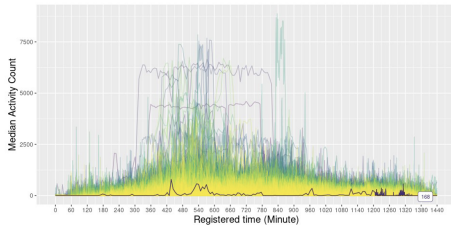
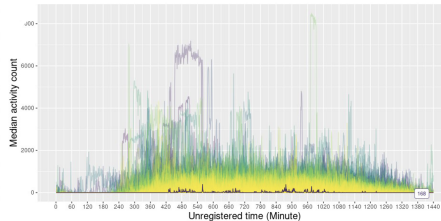
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- To use curve registration methods, a 'median day' was calculated.

Results - Curve Registration

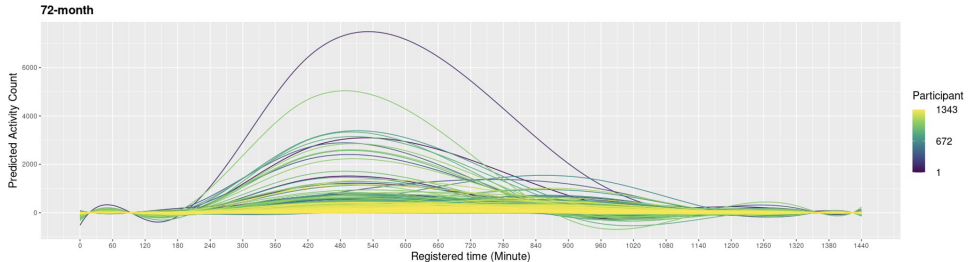
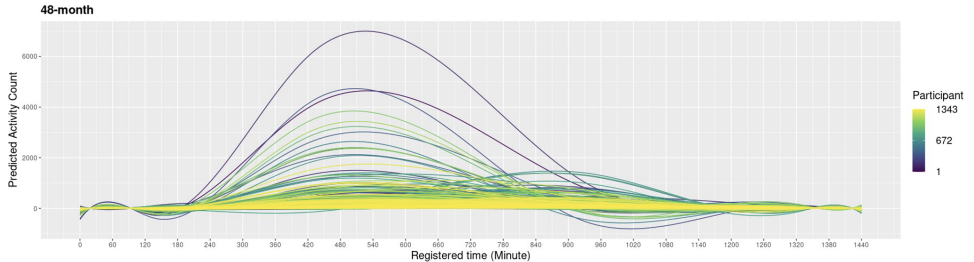
48-month Visit



72-month Visit



Results - Area Under the Registered Curve



Results - AURC across Demographic Groups of Interest

	Sample Size	48-month visit		72-month visit		Difference between visits for each subgroup
		AURC median (IQR)	Difference between subgroups	AURC median (IQR)	Difference between subgroups	
Sex			Unpaired		Unpaired	Paired
Male	616	53.34 (50.51)		46.52 (49.72)		2.3×10^{-5}
Female	721	49.76 (41.58)	$^{1}7.9 \times 10^{-3}$	45.29 (37.07)	$^{1}0.26$	3.23×10^{-5}
Race						
White	1152	52.06 (46.94)		46.61 (43.74)		2.51×10^{-9}
Non-White	185	45.72 (37.81)	$^{1}1.78 \times 10^{-4}$	42.02 (31.99)	$^{1}0.01$	0.4
Age						
< 64 at 48-month	631	48.22 (42.96)		52.42 (47.01)		1.38×10^{-7}
≥ 64 at 48-month	706	56.14 (48.45)	$^{1}9.22 \times 10^{-7}$	41.13 (34.95)	$^{1}1.28 \times 10^{-8}$	1.32×10^{-3}
Subcohort						
Progression	324	48.06 (46.27)		43.75 (35.45)		8.63×10^{-5}
Incidence	1013	52.16 (45.79)	$^{1}0.04$	47.24 (43.94)	$^{1}0.03$	4.08×10^{-6}
BMI						
Underweight and healthy weight	351	55.42 (52.11)		53.76 (46.83)		0.027
Overweight	523	54.29 (45.22)		46.24 (46.59)		8.3×10^{-5}
Obese	461	45.42 (37.93)	$^{2}6.4 \times 10^{-8}$	40.76 (34.18)	$^{2}2.64 \times 10^{-8}$	8.38×10^{-5}

The paired test (Wilcoxon signed-rank / Mann-Whitney U-test) assesses whether there is a statistically significant difference between each subgroup across the two visits, while the unpaired tests assess whether there is a statistically significant difference in AURC between the different subgroups at each visit.

BMI was categorized according to the CDC's adult BMI guidelines. The specific test used for each value is indicated by a superscript number alongside it:

¹ Unpaired Mann-Whitney U-test

² Kruskal-Wallis rank sum test

The Bonferroni correction was applied, and significant p -values are bolded ($\alpha = 2.38 \times 10^{-3}$).

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- We found that the studied OAI population presents low levels of PA.
- Found no association between AURC and long-term OA outcomes (symptomatic or structural).
- Genetic loci and predicted gene expression levels were found to be associated with AURC.

P3 - A Novel Scalar-on-Function Colocalization Method

What is colocalization?

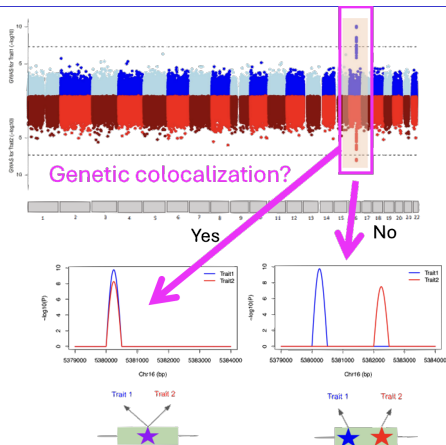
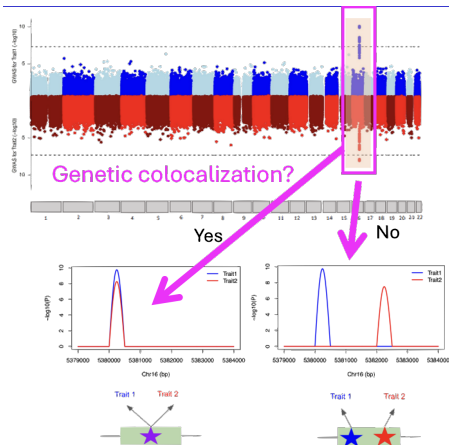


Figure adapted from Zang et al, 2024, Osteoarthritis and Cartilage Open [accepted]

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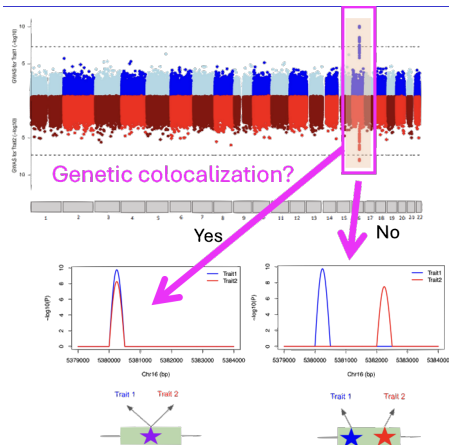


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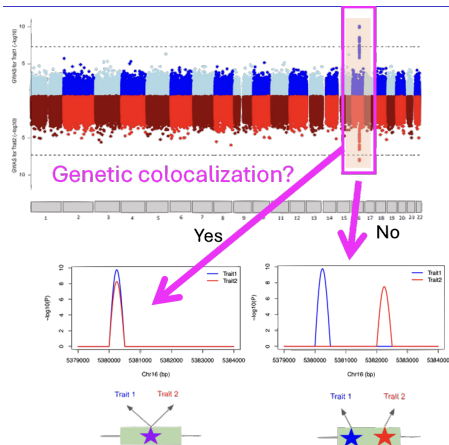


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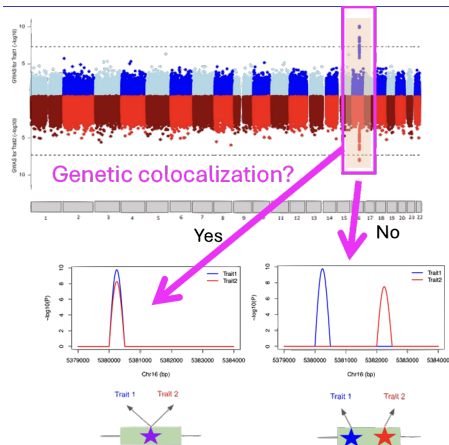


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- Colocalization methods to combined analysis of OA outcomes and miRNA levels can shed light on potential regulatory mechanisms.
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Objective: To develop a multi-trait multi-variant colocalization approach with mixed trait distributions using individual-level data.

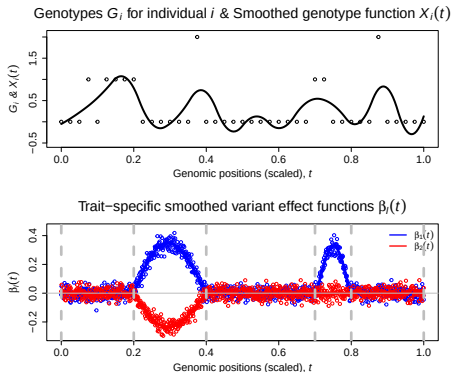
FunColoc - Model Formulation

Functional regression model for colocalization analysis

- Let y_{il} be the l th trait for the i th individual and $X_i(t)$ a continuous genotype function, which are related by the following scalar-on-function regression model

$$g_l^{-1}(E[y_{il}]) = \alpha_{0l} + \int_0^1 X_i(t)\beta_l(t)dt$$

- $g_l^{-1}(\cdot)$: trait-specific inverse link function;
- α_{0l} : intercept for trait l ;
- $\beta_l(t)$: unknown continuous genetic effect function;
- $X_i(t)$ is a smoothed function based on $\mathbf{G}_i = (G_{i1}, \dots, G_{iM})'$ genotypes.



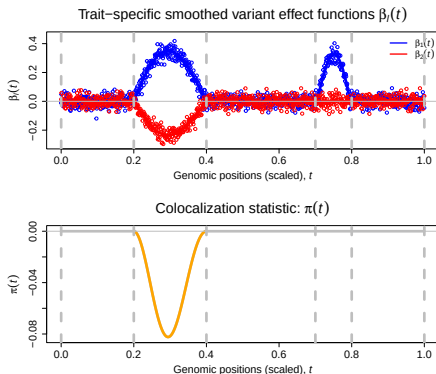
FunColoc - Model Formulation

Functional regression model for colocalization analysis

- For two traits, $l = 1, 2$ and letting $\pi(t) = \beta_1(t)\beta_2(t)$, we are interested in testing

$$H_0 : \pi(t) = 0 \text{ vs. } H_1 : \pi(t) \neq 0.$$

- Under H_0 (No colocalization)
 - $\beta_1(t) = 0$ AND $\beta_2(t) = 0$, OR
 - $\beta_1(t) = 0$ AND $\beta_2(t) \neq 0$, OR
 - $\beta_1(t) \neq 0$ AND $\beta_2(t) = 0$
- Under H_1 (Colocalization)
 - $\beta_1(t) \neq 0$ AND $\beta_2(t) \neq 0$

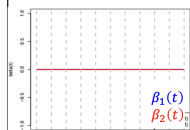


- Estimation and inference in FunColoc follows a two-step procedure using a (B-spline) basis expansion representation for $\int_0^1 X_i(t)\beta_l(t)dt$ and a multivariate covariance generalized linear model.
- Both local and global, i.e., locus-level, colocalization tests can be derived.
- Composite null hypothesis evaluated under empirical approaches (permutation and adaptive bootstrap).

Simulation Settings

Scenarios #1 to #3 under H_0 (no colocalization)

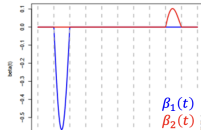
Scenario #1 (H_{00})



Scenario #2 (H_{01})

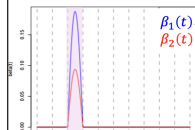


Scenario #3 (H_{01}, H_{02})

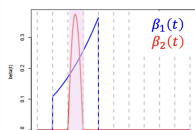


Scenarios under H_1 (colocalization)

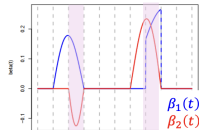
Scenario #4



Scenario #5



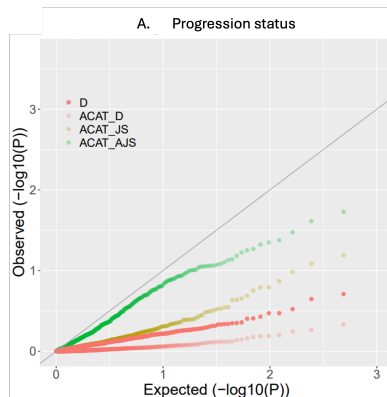
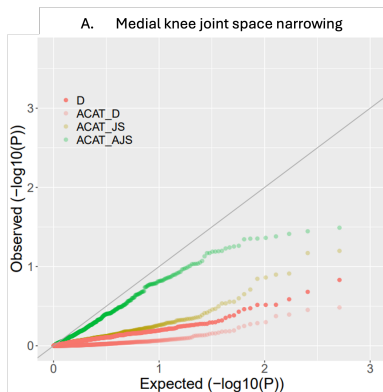
Scenario #6



- Under H_0 :
 - Controlled ($<5\%$) in all 3 scenarios.
 - Conservative when $\beta_1(t) = 0$ AND $\beta_2(t) = 0$ (Scenario 1).
- Under H_1 :
 - Increased power when both variant effect functions fully overlap (Scenario 4).

Application Results

$N = 303$ individuals with genotyping (Omni-Quad 2.5M), microRNA-seq (Illumina NextSeq550) & X-ray data.



Summary

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Summary

- **Flexible**: for combined analysis of traits with different types of distributions (e.g., normal/zero-inflated)
- Smoothed functions for genotypes & variant effects depend on variant positions
→ account for potential **non-linear genetic effects** (though parameter tuning is necessary).
- Colocalization testing:
 - FunColoc detects colocalization even in scenarios where genetic effect functions for both traits do not fully overlap
 - use of empirical approaches to account for the composite nature of H_0
 - conservative in scenarios with no genetic effects across both traits

Ongoing and Future Work

- P1
 - Analyze multiple traits to further characterize latent classes/variables (mixed data models).
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- P3
 - Implement FunColoc to more than two traits.
 - Integrate multi-omic data in a more strategic cost-efficient manner (e.g., two-phase studies)

Acknowledgements

- Trainees:
 - Reshani Abayasekara (P2)
 - Myriam Brossard (P3)
- Collaborators:
 - Ruth Fuentes-Garcia & Lizbeth Naranjo (UNAM, MX)
 - S. Amanda Ali (Henry Ford Health, US), Jackie L. Whittaker (UBC, CA).
- This research was possible in part thanks to the support of the Natural Sciences and Engineering Research Council of Canada and the Government of Canada's New Frontiers in Research Fund (NFRF) Exploration.

